

Assignment 5: Data Visualization

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Spring 2026

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
 2. Change “Student Name” on line 3 (above) with your name.
 3. Work through the steps, **creating code and output** that fulfill each instruction.
 4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
 5. Be sure to **answer the questions** in this assignment document.
 6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
-

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

#1

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026
```

```
library(cowplot)
```

```
##  
## Attaching package: 'cowplot'  
##  
## The following object is masked from 'package:lubridate':  
##  
## stamp
```

```
#2  
getwd()
```

```
## [1] "C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026"
```

```
here()
```

```
## [1] "C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026"
```

```
NTL.LTR <-read.csv(file=here('Data/Processed/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv'),  
NEON.NIWO <-read.csv(file=here('Data/Processed/NEON_NIWO_Litter_mass_trap_Processed.csv'), stringsAsFac  
str(NTL.LTR)
```

```
## 'data.frame': 24019 obs. of 15 variables:  
## $ lakename : Factor w/ 2 levels "Paul Lake","Peter Lake": 1 1 1 1 1 1 1 1 1 1 ...  
## $ year4 : int 1984 1984 1984 1984 1984 1984 1984 1984 1984 1984 ...  
## $ daynum : int 148 148 148 148 148 148 148 148 148 148 ...  
## $ month : int 5 5 5 5 5 5 5 5 5 5 ...  
## $ sampledate : Factor w/ 967 levels "1984-05-27","1984-05-28",...: 1 1 1 1 1 1 1 1 1 1 ...  
## $ depth : num 0 0.25 0.5 0.75 1 1.5 2 3 4 5 ...  
## $ temperature_C : num 14.5 NA NA NA 14.5 NA 14.2 11 7 6.1 ...  
## $ dissolvedOxygen: num 9.5 NA NA NA 8.8 NA 8.6 11.5 11.9 2.5 ...  
## $ irradianceWater: num 1750 1550 1150 975 870 610 420 220 100 34 ...  
## $ irradianceDeck : num 1620 1620 1620 1620 1620 1620 1620 1620 1620 1620 ...  
## $ tn_ug : num NA NA NA NA NA NA NA NA NA NA ...  
## $ tp_ug : num NA NA NA NA NA NA NA NA NA NA ...  
## $ nh34 : num NA NA NA NA NA NA NA NA NA NA ...  
## $ no23 : num NA NA NA NA NA NA NA NA NA NA ...  
## $ po4 : num NA NA NA NA NA NA NA NA NA NA ...
```

```
str(NEON.NIWO)
```

```
## 'data.frame': 1692 obs. of 13 variables:  
## $ plotID : Factor w/ 12 levels "NIWO_040","NIWO_041",...: 9 8 9 11 7 7 4 4 4 4 ...
```

```
## $ trapID      : Factor w/ 15 levels "NIWO_040_139",...: 11 10 11 13 9 9 5 5 5 5 ...
## $ collectDate : Factor w/ 24 levels "2016-06-16","2016-07-14",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ functionalGroup : Factor w/ 8 levels "Flowers","Leaves",...: 6 5 8 6 4 2 2 6 7 8 ...
## $ dryMass       : num  0 0.27 0.12 0 1.11 0 0 0 0.07 0.02 ...
## $ qaDryMass     : Factor w/ 2 levels "N","Y": 1 1 1 1 2 1 1 1 1 1 ...
## $ subplotID    : int   31 41 31 32 32 32 40 40 40 40 ...
## $ decimalLatitude : num   40.1 40 40.1 40 40 ...
## $ decimalLongitude: num  -106 -106 -106 -106 -106 ...
## $ elevation     : num   3477 3413 3477 3373 3446 ...
## $ nlcdClass      : Factor w/ 3 levels "evergreenForest",...: 3 1 3 1 3 3 2 2 2 2 ...
## $ plotType       : Factor w/ 1 level "tower": 1 1 1 1 1 1 1 1 1 1 ...
## $ geodeticDatum  : Factor w/ 1 level "WGS84": 1 1 1 1 1 1 1 1 1 1 ...
```

```
NTL.LTR$sampleddate <-ymd(NTL.LTR$sampleddate)

NTL.LTR$sampleddate <- as.Date(NTL.LTR$sampleddate, format = "%Y-%m-%d")
NEON.NIWO$collectDate <-as.Date(NEON.NIWO$collectDate, format = "%Y-%m-%d")
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3

myfirsttheme <- theme_classic(base_size = 12)+
  theme(
    axis.text = element_text(family="Palatino", color="#0A1A2F"),
    legend.position="top",
    plot.background = element_rect(fill="#FAFAFA", color=NA), plot.title = element_text(color="#0A1A2F"))

theme_set(myfirsttheme) #setting this theme as default
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

#4

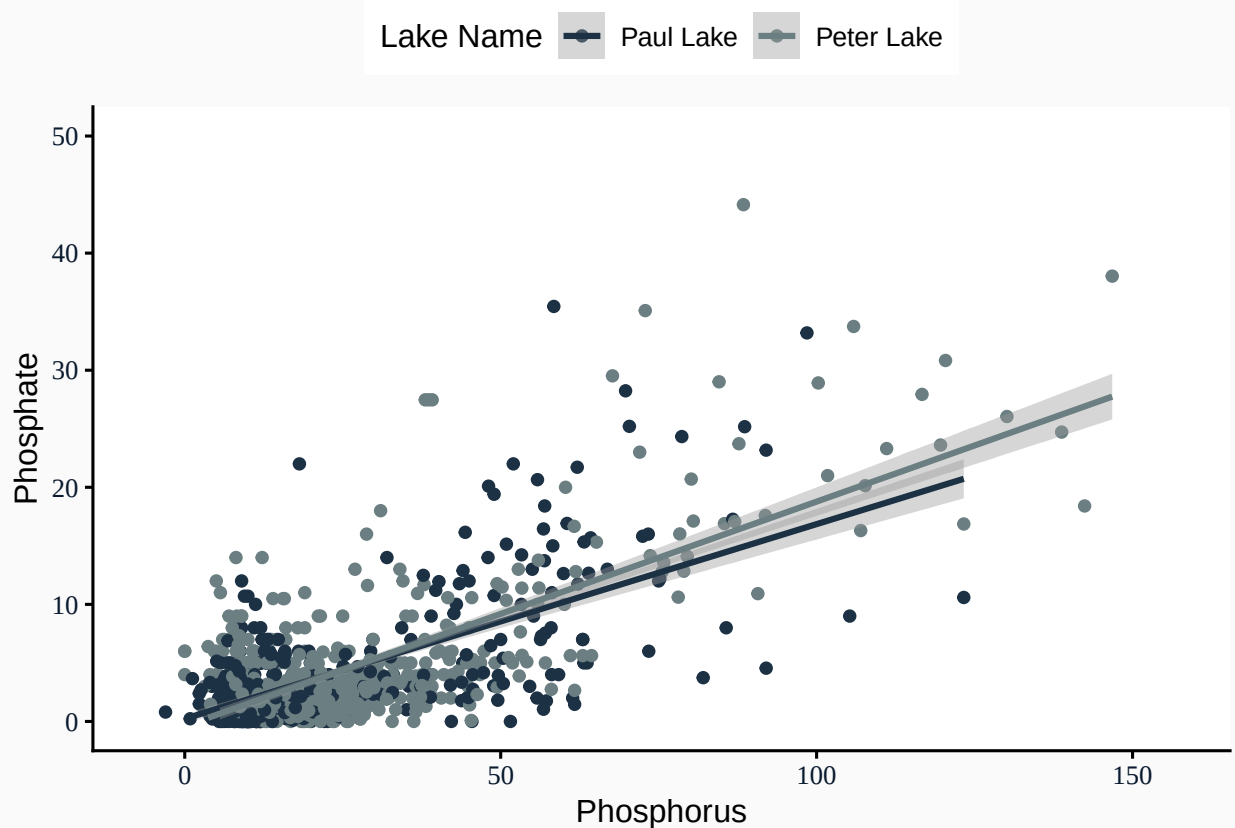
```
PP04_Plot <- ggplot(NTL.LTR, aes(x=tp_ug, y=po4, color=lakename)) +  
  geom_point()+  
  geom_smooth(method='lm')+  
  ylim(0,50)+  
  labs(  
    color="Lake Name",  
    x="Phosphorus",  
    y="Phosphate"  
  )+  
  scale_color_manual(values = c(  
    "Paul Lake" = "#1C3144",  
    "Peter Lake" = "#6B7F82"  
  ))  
  
print(PP04_Plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 22958 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

```
## Warning: Removed 22958 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range  
## ('geom_smooth()').
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned. Show all months, even those where no data was collected.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis text in your theme to `element_blank()` removes the axis text (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different relative sizes when combined using `cowplot`.

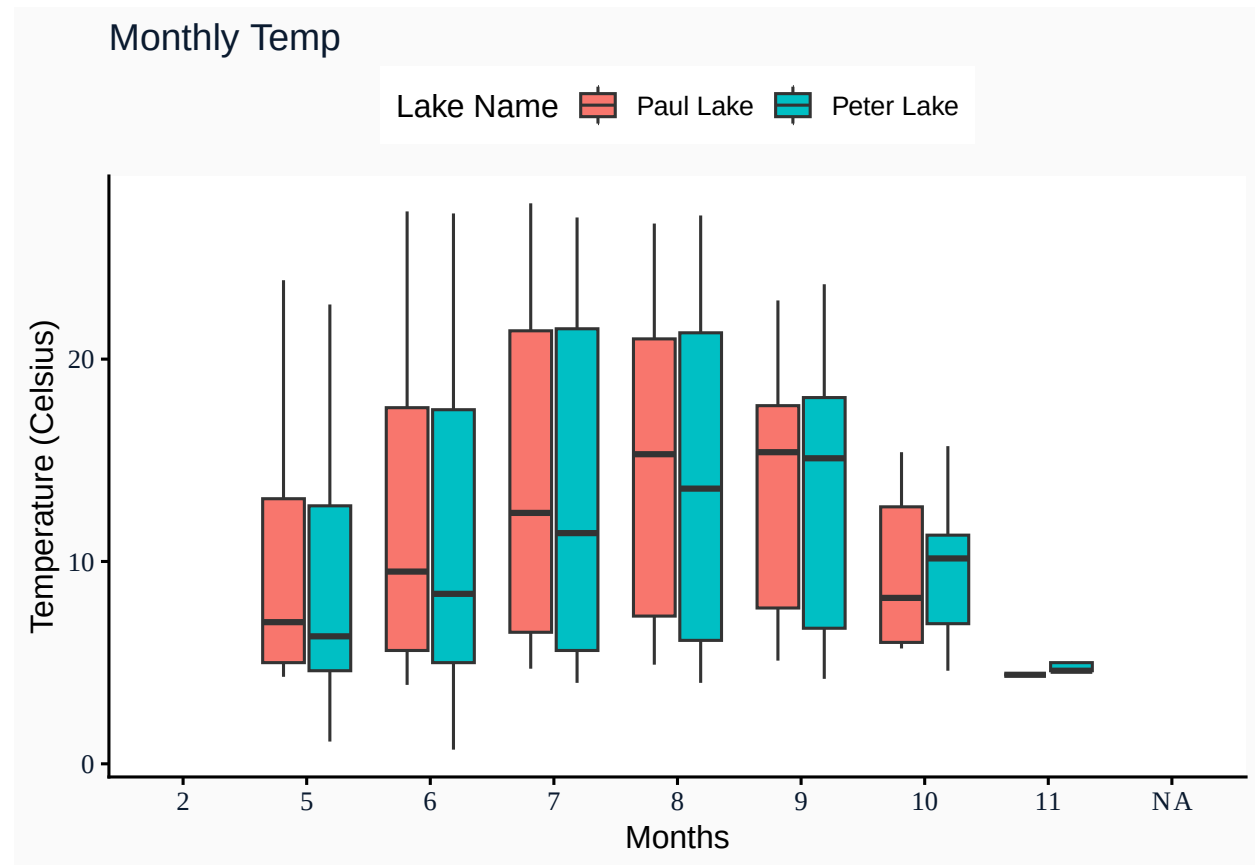
```
#5

temp_boxlegend <- ggplot(NTL.LTR, aes(
  x =factor(month),
  y=temperature_C,
  fill=lakename),
)+
labs(
  fill="Lake Name", #this is where we change the legend title since we extract here
  title ="Monthly Temp",
  x="Months",
  y="Temperature (Celsius)")
theme(
  plot.title = element_text(margin = margin(b=3))
)+
```

```
geom_boxplot()
```

```
print(temp_boxlegend)
```

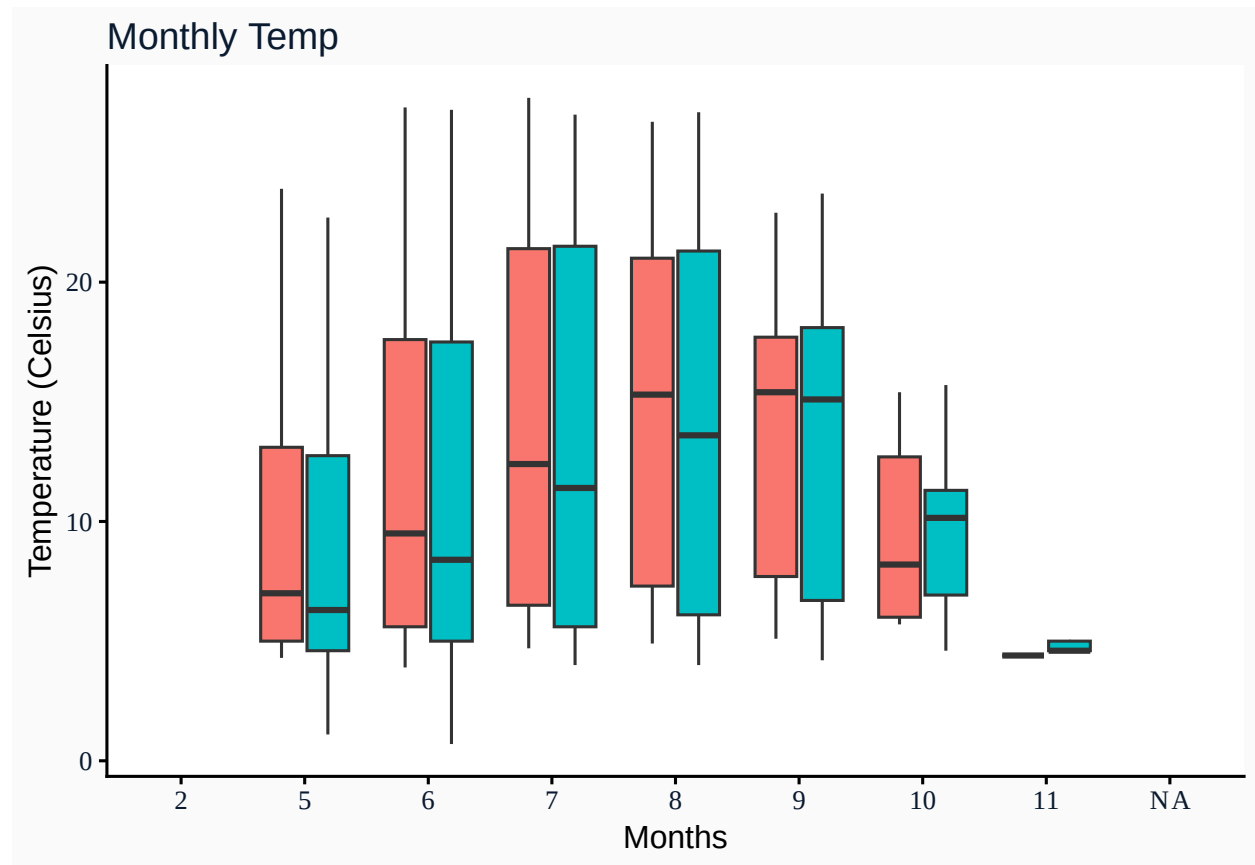
```
## Warning: Removed 4577 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```



```
temp_box <- ggplot(NTL.LTR, aes(  
  x =factor(month),  
  y=temperature_C,  
  fill=lakename),  
)+  
labs(  
  title ="Monthly Temp",  
  x="Months",  
  y="Temperature (Celsius)" )+  
theme(  
  legend.position = "none",  
  plot.title = element_text(margin = margin(b=3))  
)+  
geom_boxplot()  
  
print(temp_box)
```

```
## Warning: Removed 4577 rows containing non-finite outside the scale range
```

```
## ('stat_boxplot()').
```



```
#trying out with sampleddate instead of month factor.ignore this. it's just me experimenting.
```

```
#ignore
```

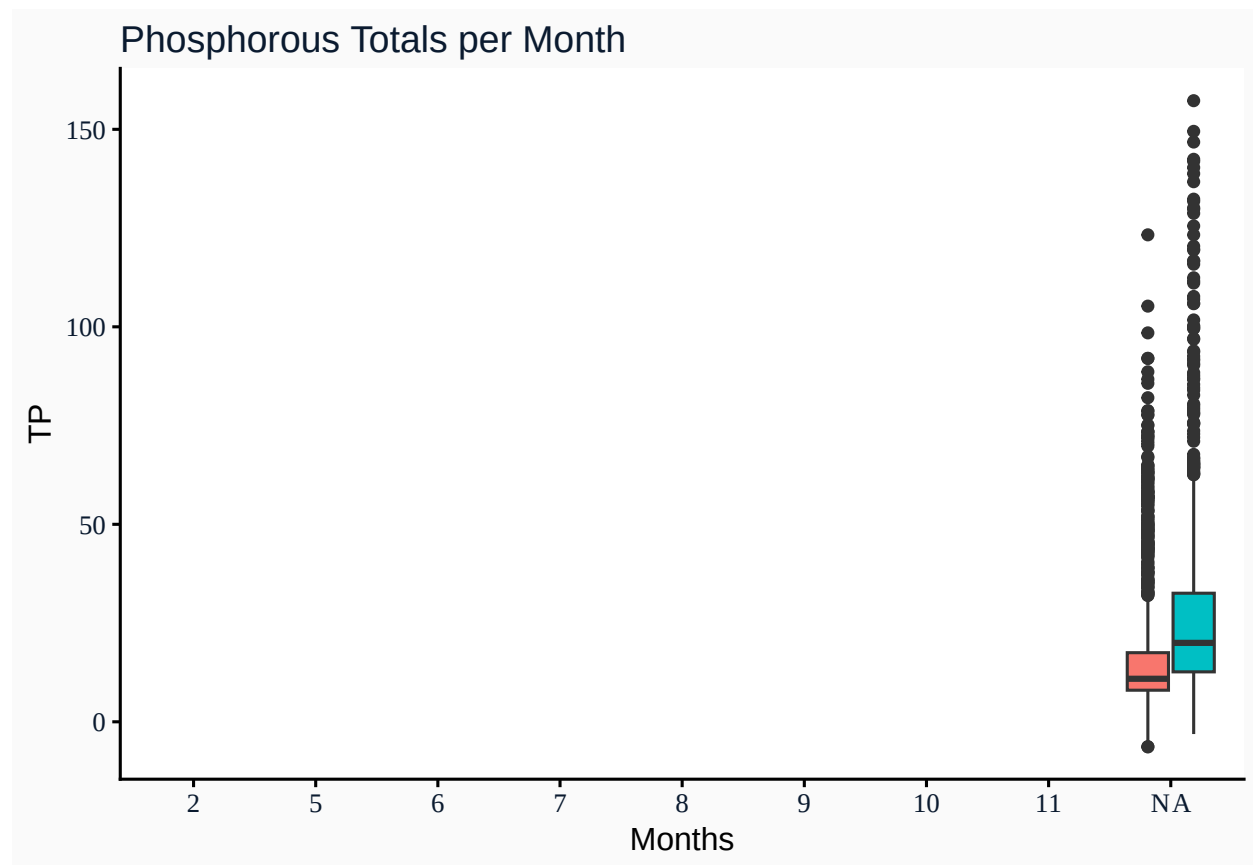
```
tempboxsampledate <-ggplot(NTL.LTR, aes(  
  x=factor(format(NTL.LTR$sampledate, "%Y")),  
  y=temperature_C)  
) +  
  geom_boxplot() +  
  labs(  
    title = "Temp Data",  
    x="date",  
    y="temperature"  
  )
```

```
#print(tempboxsampledate)
```

```
tp_box <-ggplot(NTL.LTR, aes(
  x=factor(month),
  y=tp_ug,
  fill=lakename)
)+
labs(
  title = "Phosphorous Totals per Month",
  x="Months",
  y="TP")+
theme(
  legend.position = "none",
  plot.title = element_text(margin=margin(b=3))
)+
geom_boxplot()

print(tp_box)
```

```
## Warning: Removed 21740 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



```
tn_box <-ggplot(NTL.LTR, aes(
  x=factor(month),
  y=tn_ug,
  fill=lakename)
```

```

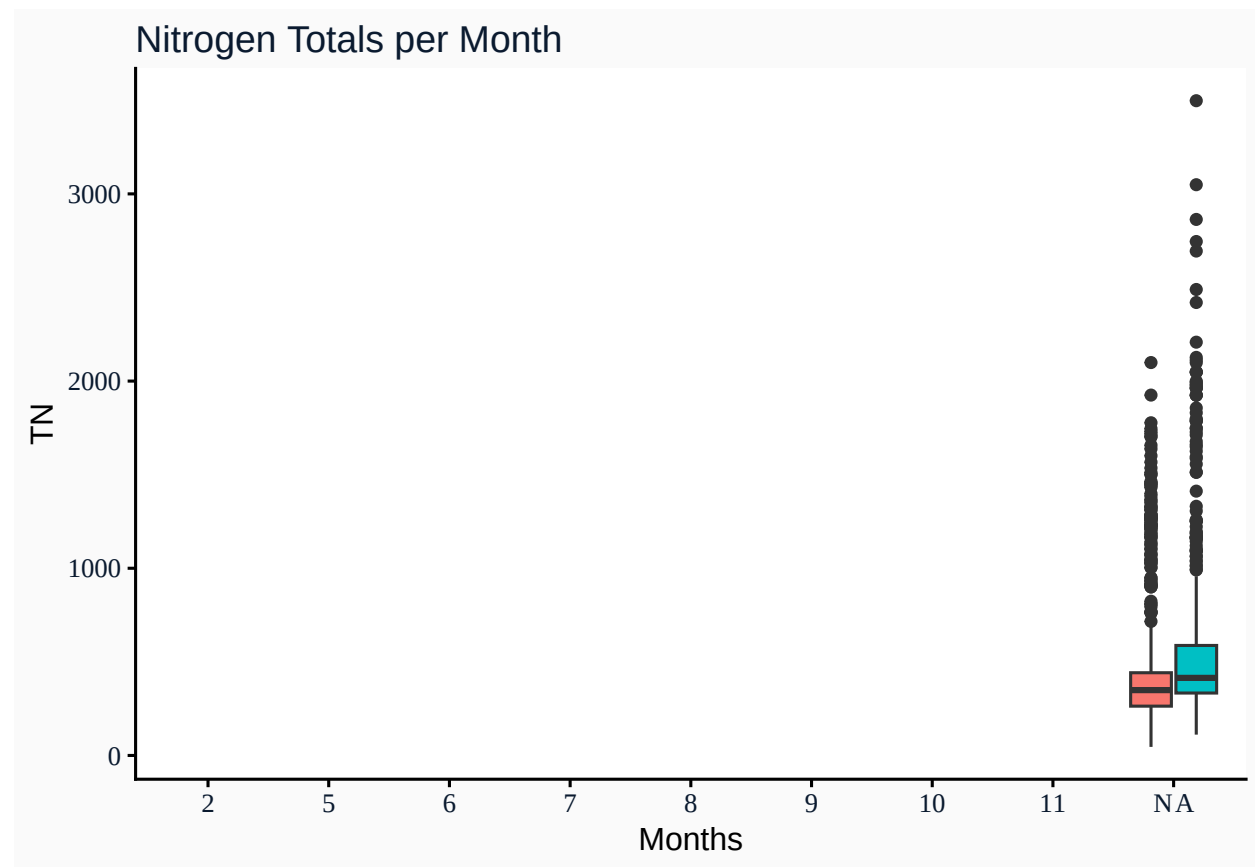
)+
labs(
  title = "Nitrogen Totals per Month",
  x="Months",
  y="TN"
)+
theme(
  legend.position = "none",
  plot.title = element_text(margin = (margin(b=3)))
)+
geom_boxplot()
print(tn_box)

```

```

## Warning: Removed 22594 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```



```

#save legend as object

grid_legend <- get_legend(
  temp_boxlegend +
  theme(legend.position = "right")
)

```

```

## Warning: Removed 4577 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

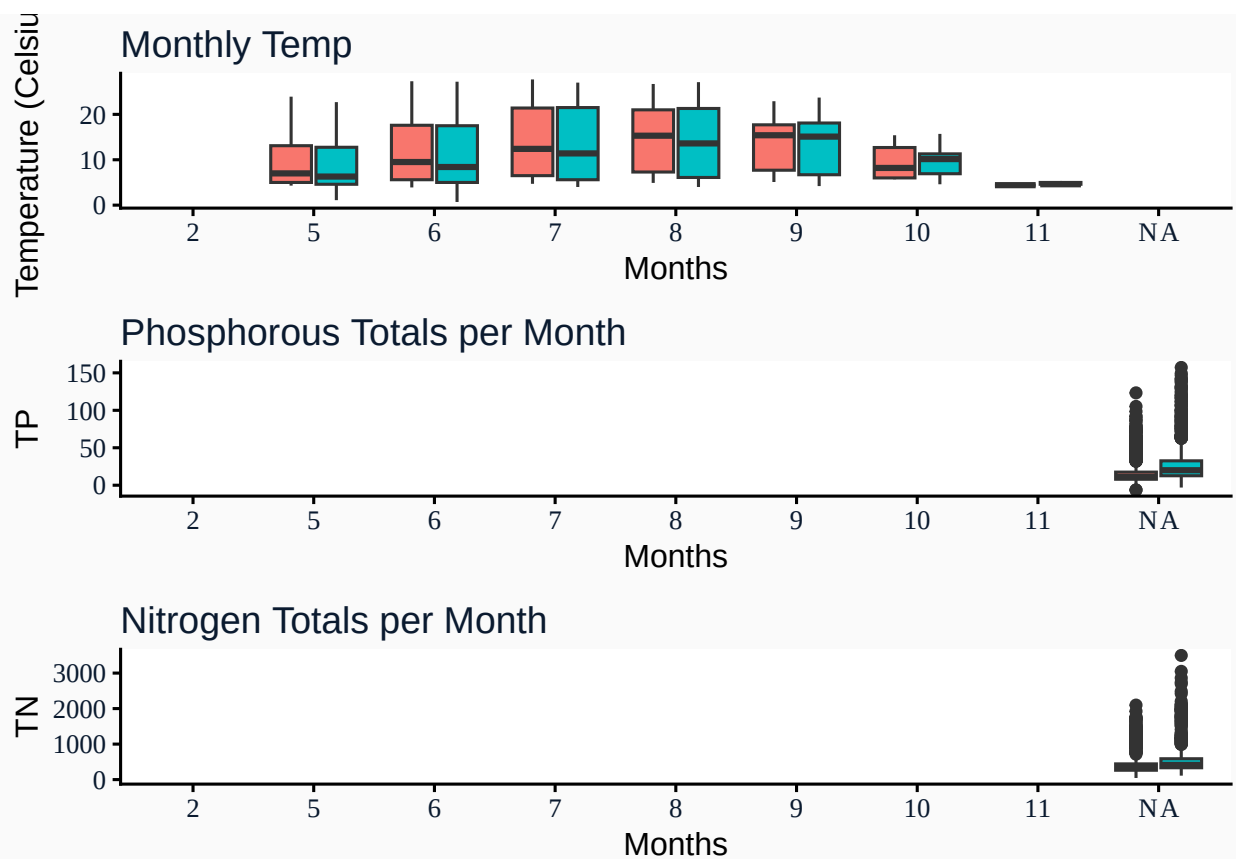
```
box_grid_aligned <- plot_grid(temp_box, tp_box, tn_box, align = 'hv', rel_widths = c(1,1,1), ncol = 1)
```

```
## Warning: Removed 4577 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 21740 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 22594 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
print(box_grid_aligned)
```

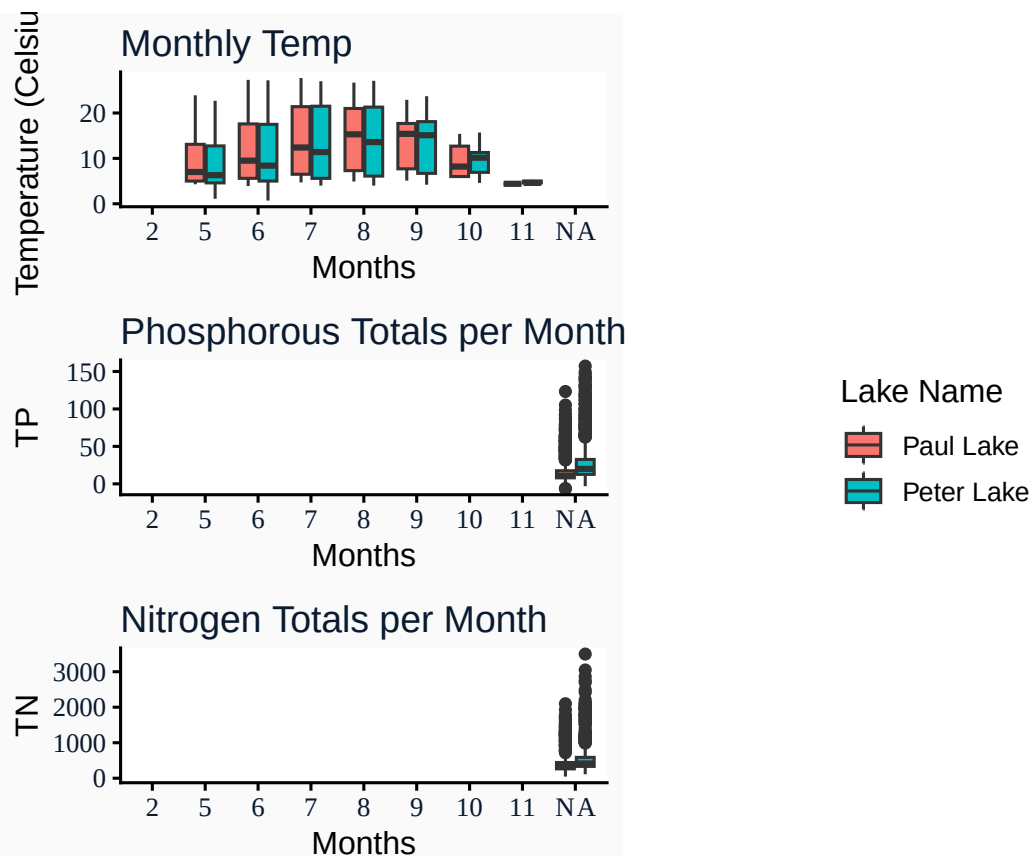


```
#reference link https://wilkelab.org/cowplot/articles/shared\_legends.html
```

```
#add legend
```

```
box_grid_legacy <- plot_grid(box_grid_aligned, grid_legend, nrow = 1)
```

```
print(box_grid_legacy)
```



Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: Don't know if I am using the wrong dataset considering all nitrogen and phosphorous data is missing a date, so there is not much to observe there other than that Peter Lake has higher totals of both and that temperatures are predictably higher in July and August.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6

suppressWarnings({ #open warning suppression

#6-a practicing via data wrangling method. see 6-b for subset method.
niwot.needles <- NEON.NIWO %>%
  filter(functionalGroup=="Needles")

niwot.needle.box <- ggplot(niwot.needles, aes(
  x= factor(collectDate),
  y=dryMass,
  fill=nlcdClass,
```

```

)))+
  labs(
    title = "Needle Dry Mass",
    x = "Date",
    y = "Dry Mass",
    fill = "NCLD Class"
  )+
  geom_boxplot()

print(niwot.needle.box)

#6-b subset method

NEON.NIWO$Year <- format(NEON.NIWO$collectDate, "%Y")
NEON.NIWO$Month <- format(NEON.NIWO$collectDate, "%m")

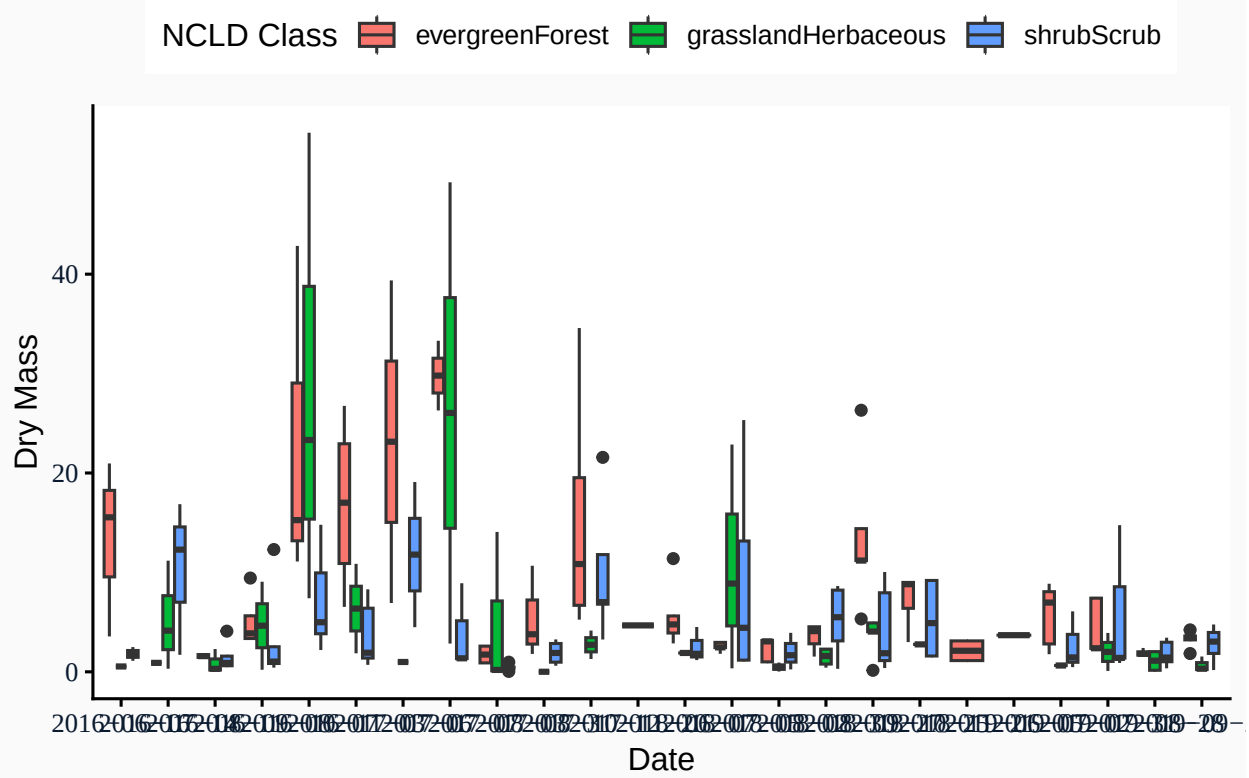
niwot.subset.box <- ggplot(subset(NEON.NIWO, functionalGroup == "Needles"),
  aes(
    x = factor(Month),
    y = dryMass,
    fill = nlcdClass
  ))+
  facet_grid(nlcdClass ~ Year,
    labeller = labeller(nlcdClass = c("evergreenForest" = "Evergreen Forest",
    "grasslandHerbaceous" = "Grassland", "shrubScrub" = "Shrub Scrub")
  ))+
  geom_boxplot()+
  labs(
    x = "Collection Month",
    y = "Dry Mass",
    fill = "NLCD Class"
  )+
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8)
  )+
  scale_fill_discrete(
    labels = c(
      "evergreenForest" = "Evergreen Forest",
      "grasslandHerbaceous" = "Grassland",
      "shrubScrub" = "Shrub Scrub"
    )
  )

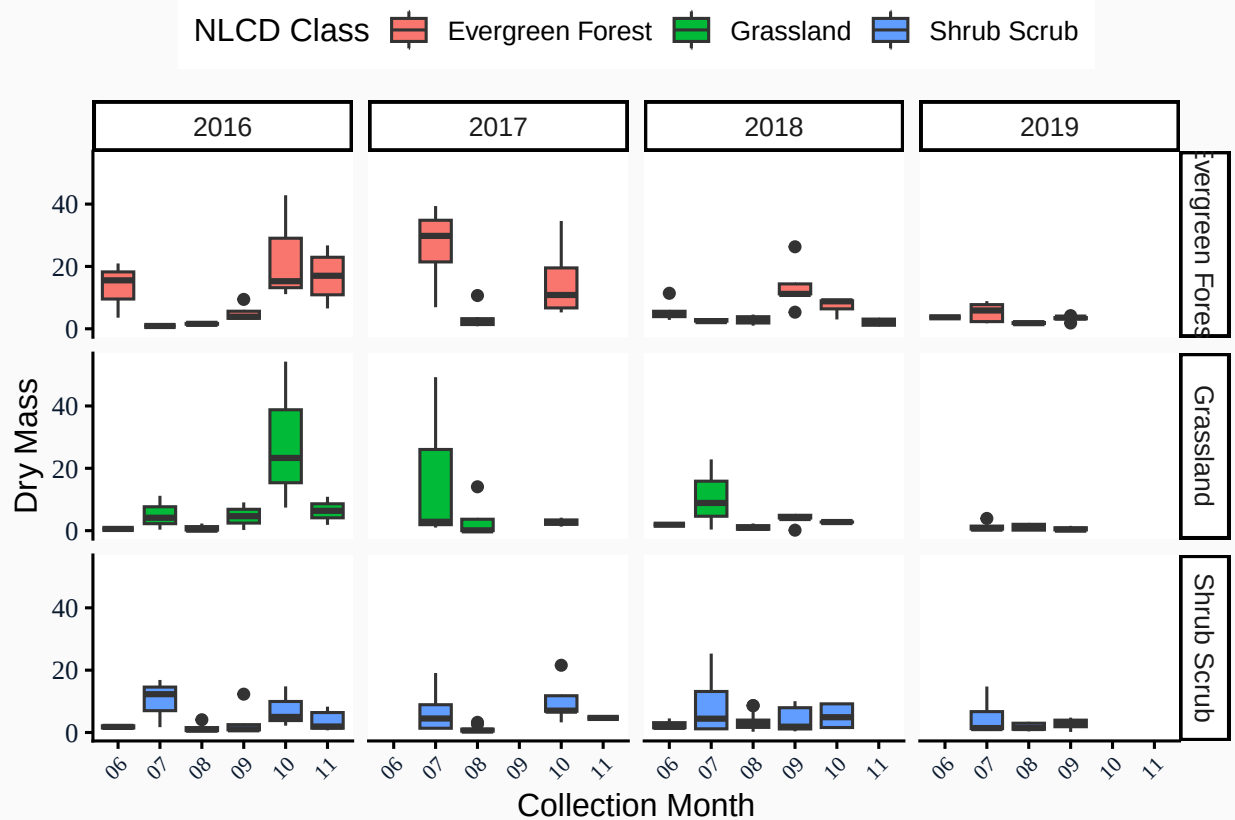
print(niwot.subset.box) #this is the best plot of the two strategies.

}) #close warning suppression

```

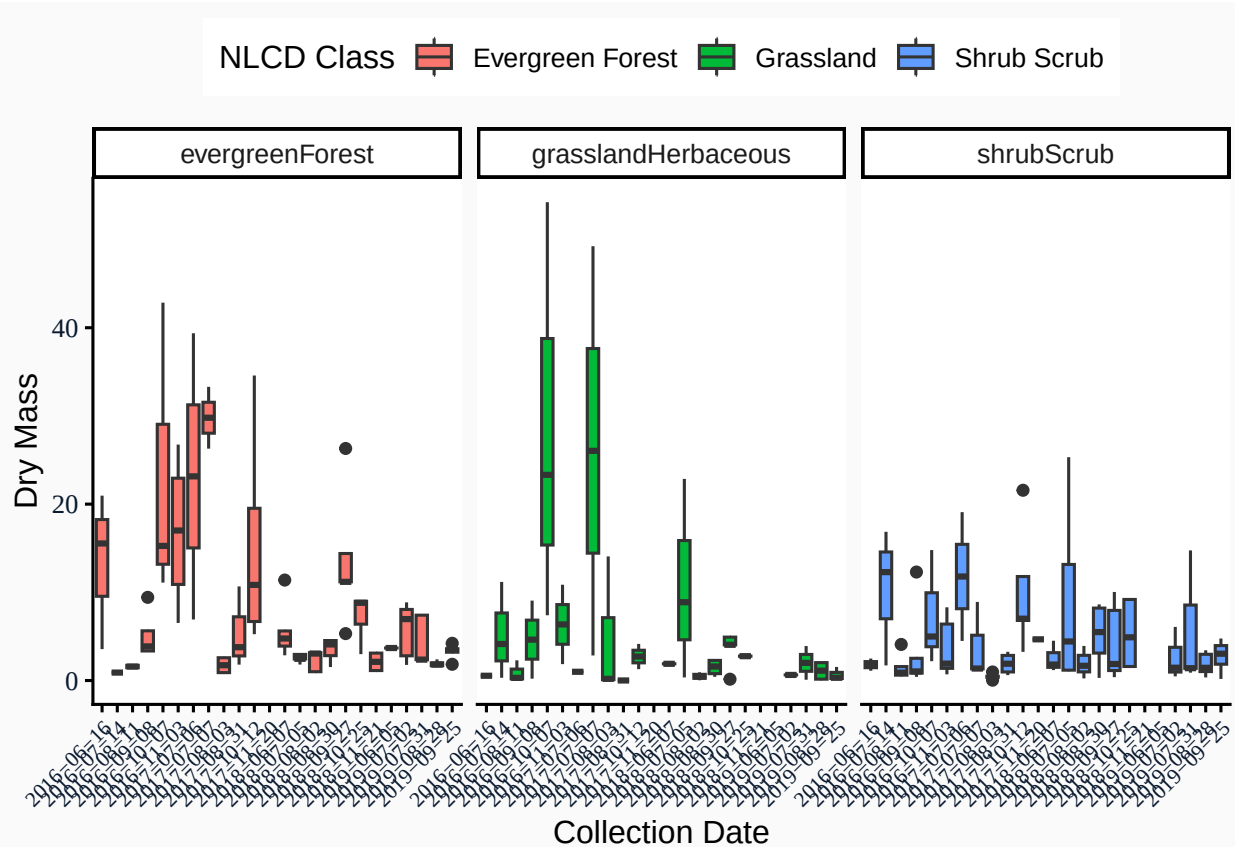
Needle Dry Mass





```
needles.face <- ggplot(niwot.needles, aes(
  x=factor(collectDate),
  y=dryMass,
  fill=nlcdClass
)) +
  labs(
    x="Collection Date",
    y="Dry Mass",
    fill="NLCD Class"
  ) + theme(
    axis.text.x = element_text(angle=45, hjust = 1, size=8)
  ) +
  geom_boxplot() +
  facet_wrap(vars(nlcdClass), nrow=1) +
  scale_fill_discrete(
    labels = c(
      "evergreenForest" = "Evergreen Forest",
      "grasslandHerbaceous" = "Grassland",
      "shrubScrub" = "Shrub Scrub"
    )
  )

print(needles.face)
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: The three-faceted plot is more effective. You can easily compare Dry Mass between months and compare between year years. Unlike the other plot, the `facet_grid` method makes the pattern of decreasing dry mass between 2016 - 2019 obvious at a glance.